



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005
& ANSI/NCSL Z540-1-1994

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CALIBRATION

Valid To: May 31, 2017

Certificate Number: 3128.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Cell Phone Test System

Parameter/Equipment	Range	CMC ^{2,4} (±)	Comments
W-CDMA ³ TRx/Performance Test System (ME7873F) –			
UL Measurement Accuracy	Max Output	0.27 dB	Signal analyzer, signal generator, power meter and power sensor, calibration receiver
(0.5 to 2) GHz	TX IM(UL)	0.27 dB	
Modulation	TX IM(IM)	0.28 dB	
	Min Output	0.28 dB	
Relative Level Measurement Accuracy			
(0.5 to 2) GHz	1 dB Step	0.01 dB	
Modulation	2 dB Step	0.01 dB	
	3 dB Step	0.01 dB	
	26 dB Step	0.20 dB	
RACH Measurement Accuracy (Absolute Level Measurement)			
(0.5 to 2) GHz	Pre1f (+5 dBm)	0.29 dB	
Modulation	Pre1f	0.31 dB	
	PreMAX	0.27 dB	
	Off Power	0.34 dB	

Parameter/Equipment	Range	CMC ^{2,4} (±)	Comments
W-CDMA ³ TRx/Performance Test System (ME7873F) – (cont)			
RACH Measurement Accuracy (Relative Level Measurement)			
(0.5 to 2) GHz Modulation	Power Difference	0.01 dB	Signal analyzer, signal generator, power meter and power sensor, calibration receiver
DPCCH Measurement Accuracy			
(0.5 to 2) GHz Modulation	8.4.4	0.30 dB	
	8.7.9	0.30 dB	
	8.7.9 (UL)	0.30 dB	
TX Spurious Measurement Accuracy (General) CW	(0.15 to 10) MHz	0.27 dB	
	(10 to 2200) MHz	0.32 dB	
	(2.2 to 4) GHz	0.61 dB	
	(4 to 12.75) GHz	1.2 dB	
TX Spurious Measurement Accuracy (Additional) CW			
GSM	(921 to 925) MHz	0.30 dB	
	(925 to 960) MHz	0.31 dB	
	(1805 to 1880) MHz	0.31 dB	
PHS	(1884.5 to 1919.6) MHz	0.30 dB	
DL Band	Band I, II, III, IV, V, VII, VIII, IX, XI; 800 MHz	0.33 dB	
RX Spurious Measurement Accuracy (General) CW	(30 to 1000) MHz	0.30 dB	
	(1 to 2.2) GHz	0.32 dB	
	(2.2 to 4) GHz	0.61 dB	
	(4 to 13) GHz	1.2 dB	
RX Spurious Measurement Accuracy (Additional) CW			
In Band	(0.5 to 2) GHz	0.33 dB	
GSM	(921 to 925) MHz	0.30 dB	
	(925 to 960) MHz	0.30 dB	
Spectrum Emission Mask CW	(0.5 to 2) GHz CW	0.27 dB	

Parameter/Equipment	Range	CMC ^{2,4} (±)	Comments
W-CDMA ³ TRx/Performance Test System (ME7873F) – (cont)			
Adjacent Channel Leakage Power Ratio (ACLR)	(0.5 to 2) GHz, Modulation/CW	0.43 dB	Signal analyzer, signal generator, power meter and power sensor, calibration receiver
Occupied Bandwidth (OBW) Modulation	(0.5 to 2) GHz	21 kHz	
Frequency Error Modulation	(0.5 to 2) GHz	0.02 Hz	
Error Vector Magnitude (EVM)	ULRMC12k	0.66 % of full scale	
(0.5 to 2) GHz Modulation	Pre1f	0.66 % of full scale	
Phase Error			
(0.5 to 2) GHz Modulation		0.23°	
Peak Code Domain Error			
(0.5 to 2) GHz Modulation		0.12 dB	
DL Output Accuracy			
(0.5 to 3) GHz Modulation			
Downlink Power	(-30 to 0) dBm (-80 to -30) dBm; Ioc (MBMS)	0.32 dB 0.37 dB	
	(-120 to -80) dBm Blocking (In-Band), Spurious	0.45 dB	
	Max Input	0.26 dB	
Ior/Ioc (TRx)			
(0.5 to 3) GHz Modulation	DPCCH_Ec/Ioc Ior/Ioc Ioc	0.09 dB 0.09 dB 0.34 dB	

Parameter/Equipment	Range	CMC ^{2, 4} (±)	Comments
W-CDMA ³ TRx/Performance Test System (ME7873F) – (cont)			
Ior/Ioc (Perf) (0.5 to 3) GHz Modulation	Ior/Ioc(Static) Ior/Ioc(Static)[$\sqrt{0.3^2(\text{Static}) + 0.3^2(\text{Static})}$] Ioc Fading Ior/Ioc[$\sqrt{0.3^2(\text{Static}) + 0.5^2(\text{Fading})}$] Ior/Ioc[$\sqrt{0.3^2(\text{Static}) + 0.5^2(\text{Fading}) + 0.5^2(\text{Fading})}$] Ior/Ioc[3000times average]	0.09 dB 0.12 dB 0.30 dB 0.09 dB 0.12 dB 0.15 dB 0.10 dB	Signal analyzer, signal generator, power meter and power sensor, calibration receiver
Ior/Ioc (RRM Fading Test Case) PropSimC2 (0.5 to 3) GHz Modulation	Ioc Ior Ior1/Ior2	0.31 dB 0.31 dB 0.09 dB	
Ior/Ioc (RRM Fading Test Case) MF6900A (0.5 to 3) GHz Modulation	Ioc Ior	0.28 dB 0.29 dB	
RRM Signal Output (0.5 to 3) GHz Modulation	Ior, Ioc, RXLEV Ior/Ioc, Ior/Ior, Ioc/Ioc Ioc/RXLEV, Ioc/RXLEV (8.7.3A), Ior/Ior (Inter Band), Ioc/Ioc (Inter Band), Ioc/RXLEV (Ior and Ioc are combined in digital baseband oSFS)	0.28 dB 0.09 dB 0.12 dB	

Parameter/Equipment	Range	CMC ^{2, 4} (±)	Comments/ Procedure/ Budget
W-CDMA ³ TRx/Performance Test System (ME7873F) – (cont)			
CW Interference	RXLEV/RXLEV	0.39 dB	Signal analyzer, signal generator, power meter and power sensor, calibration receiver
	Blocking: (10 to 3000) MHz	0.26 dB	
	Blocking: (3 to 8) GHz	0.54 dB	
	Blocking: (8 to 13) GHz	1.1 dB	
	RX IM	0.26 dB	
	TX IM	0.26 dB	
Modulation Interference	SG2->CpDSG->Isolator->Antenna	0.29 dB	
(0.5 to 3) GHz Modulation	SG2->Isolator->Antenna	0.29 dB	
	Intermodulation Characteristics	0.26 dB	
GSM Interference			
(0.5 to 3) GHz Modulation	Blocking Characteristics (Narrow Band)	0.29 dB	
	Intermodulation Characteristics (Narrow Band)	0.26 dB	

Parameter/Equipment	Range	CMC ^{2, 4} (±)	Comments/ Procedure/ Budget
LTE/A RF Conformance ³ Test System (ME78xx ⁷) –			
UL Measurement Accuracy			
(0.5 to 3) GHz Modulation, CW	Max Output Min Output Off Power TX IM (UL) TX IM (IM) Carrier Aggregation	0.37 dB 0.26 dB	Signal generator, power meter, power sensor, signal analyzer
UL Measurement Accuracy			
Freq. ≤ 3 GHz 3 GHz < Freq. ≤ 4 GHz	Max Output Max Output	0.30 dB 0.35 dB	
Frequency Error	(0.5 to 3) GHz Modulation	0.02 Hz	
Error Vector Magnitude (EVM)	(0.5 to 3) GHz Modulation	0.21 % of full scale	
Origin Offset	(0.5 to 3) GHz Modulation	0.44 dB	
In Band Emission In Band Emission for CA	(0.5 to 3) GHz Modulation	0.27 dB 0.29 dB	
Occupied Bandwidth (OBW)	(0.5 to 3) GHz Modulation	0.01 kHz	
Spectrum Emission Mask	(0.5 to 3) GHz, CW	0.27 dB	
Adjacent Channel Leakage power Ratio (ACLR)	(0.5 to 3) GHz Modulation, CW	0.30 dB	
Spectrum Flatness	(0.5 to 3) GHz Modulation	0.60 dB	
Spurious, CW	(10 to 30) MHz (30 to 4000) MHz (4 to 13) GHz	0.29 dB 0.61 dB 1.2 dB	

Parameter/Equipment	Range	CMC ^{2, 4} ($\pm$)	Comments
LTE/A RF Conformance ³ Test System (ME78*) (cont)			
DL Output Accuracy	Bandwidth: 1.4 MHz	0.35 dB	Signal generator, power meter, power sensor, signal analyzer
(0.5 to 3) GHz Modulation	Bandwidth: 20 MHz	0.33 dB	
DL Output Accuracy (RRM)	LTE absolute	0.29 dB	
(0.5 to 3) GHz Modulation, CW	AWGN absolute		
	W-CDMA absolute	0.30 dB	
	GSM absolute	0.27 dB	
	Memory synthesizing function output	0.26 dB	
	CDMA2000 absolute	0.29 dB	
	LTE / LTE relative (InterFreq.), LTE / W-CDMA relative, LTE / CDMA2000 relative, CDMA2000 / AWGN relative	0.41 dB	
	LTE / GSM relative	0.39 dB	
	DC-HSDPA Ior/Ioc	0.10 dB	
	TDS-CDMA Ior/Ioc	0.10 dB	
DL Output Accuracy			
Freq. $\leq$ 3 GHz	LA Absolute	0.30 dB	
3 GHz $<$ Freq. $\leq$ 4GHz	LA Absolute	0.50 dB	
CW Interference, CW			
Out of Band	(10 to 3000) MHz	0.26 dB	
	(3 to 4) GHz	0.54 dB	
Near Band	(4 to 13) GHz	1.1 dB	
Modulation Interference	(680 to 2700) MHz; (-46 to -15) dBm	0.26 dB	
(0.5 to 3) GHz Modulation	LTE, MediaFLO, ATSC	0.35 dB	
UL Relative Power Measurement	(0.5 to 3) GHz Modulation	0.23 dB	

II. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC ² ($\pm$)	Comments
DC Voltage – Generate	1 mV 10 mV (22 to 220) mV (0.22 to 2.2) V (2.2 to 11) V (11 to 220) V (220 to 1 000) V	2.3 μ V 2.3 μ V 8.1 μ V/V + 2.3 μ V 7.1 μ V/V + 2.4 μ V 8.1 μ V/V + 4.1 μ V 8.1 μ V/V + 81 μ V 9.1 μ V/V + 510 μ V	Calibrator
DC Voltage – Measure	1 mV (10 to 100) mV (0.1 to 1) V (1 to 10) V (10 to 100) V (100 to 1 000) V	2.2 μ V 5.8 μ V/V + 2.2 μ V 4.7 μ V/V + 2.2 μ V 3.9 μ V/V + 1.9 μ V ⁶ 5.8 μ V/V + 29 μ V ⁶ 18 μ V/V + 96 μ V ⁶	Multimeter
DC Current – Generate	(22 to 220) μ A (0.22 to 2.2) mA (2.2 to 22) mA (22 to 220) mA (0.22 to 1) A	51 μ A/A + 8.1 nA 51 μ A/A + 8.1 nA 51 μ A/A + 0.081 μ A 61 μ A/A + 0.81 μ A 81 μ A/A + 26 μ A	Calibrator
DC Current – Measure	(10 to 100) μ A (0.1 to 1) mA (1 to 10) mA (10 to 100) mA (0.1 to 1) A	24 μ A/A + 0.93 nA 20 μ A/A + 4.8 nA ⁶ 20 μ A/A + 0.048 μ A ⁶ 34 μ A/A + 0.48 μ A ⁶ 0.011 % + 9.6 μ A ⁶	Multimeter
Resistance – Generate, Fixed Points	1 Ω 10 Ω 100 Ω 1 k Ω 10 k Ω 100 k Ω 1 M Ω 10 M Ω 100 M Ω	0.24 m Ω 0.36 m Ω 2.8 m Ω 14 m Ω 130 m Ω 1.5 Ω 21 Ω 410 Ω 12 k Ω	Calibrator

Parameter/Equipment	Range	CMC ² (±)	Comments
Resistance – Measure	(1 to 10) Ω (10 to 100) Ω (0.1 to 1) kΩ (1 to 10) kΩ (10 to 100) kΩ (0.1 to 1) MΩ (1 to 10) MΩ (10 to 100) MΩ (0.1 to 1) GΩ	18 μΩ/Ω + 230 μΩ 14 μΩ/Ω + 2.3 mΩ 9.6 μΩ/Ω + 1.9 mΩ ⁶ 9.6 μΩ/Ω + 19 mΩ ⁶ 9.6 μΩ/Ω + 60 mΩ ⁶ 15 μΩ/Ω + 2.0 Ω ⁶ 48 μΩ/Ω + 96 Ω ⁶ 0.048 % + 960 Ω ⁶ 0.49 % + 9.6 kΩ ⁶	Multimeter

Parameter/Equipment	Frequency	CMC ^{2,5} (±)	Comments
AC Voltage – Generate			
(22 to 220) mV	(20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.022 % + 8.0 μV 0.012 % + 8.0 μV 0.033 % + 8.0 μV 0.086 % + 25 μV 0.12 % + 25 μV 0.18 % + 35 μV 0.35 % + 80 μV	Calibrator
(0.22 to 2.2) V	(20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.017 % + 53 μV 76 μV/V + 24 μV 0.013 % + 29 μV 0.026 % + 74 μV 0.044 % + 180 μV 0.11 % + 370 μV 0.23 % + 860 μV	
(2.2 to 22) V	(20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.017 % + 260 μV 76 μV/V + 61 μV 0.013 % + 170 μV 0.026 % + 360 μV 0.051 % + 1.6 mV 0.13 % + 4.4 mV 0.28 % + 8.6 mV	
(22 to 220) V	(20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz	0.017 % + 2.6 mV 81 μV/V + 810 μV 0.023 % + 3.6 mV 0.051 % + 8.1 mV	
(220 to 700) V	50 Hz to 1 kHz	81 μV/V + 3.6 mV	

Parameter/Equipment	Frequency	CMC ^{2,5} ($\pm$)	Comments
AC Voltage – Measure			
(10 to 100) mV	(20 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz 300 kHz to 1 MHz	0.01 % + 4.7 μ V 0.01 % + 2.4 μ V 0.018 % + 2.4 μ V 0.029 % + 2.0 μ V ⁶ 0.077 % + 2.0 μ V ⁶ 0.29 % + 9.6 μ V ⁶ 0.96 % + 9.6 μ V ⁶	Multimeter
(0.1 to 1) V	(20 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz 300 kHz to 1 MHz	81 μ V/V + 47 μ V 67 μ V/V + 20 μ V ⁶ 0.014 % + 20 μ V ⁶ 0.029 % + 20 μ V ⁶ 0.077 % + 20 μ V ⁶ 0.29 % + 96 μ V ⁶ 0.96 % + 96 μ V ⁶	
(1 to 10) V	(20 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz 300 kHz to 1 MHz	81 μ V/V + 470 μ V 67 μ V/V + 200 μ V ⁶ 0.014 % + 200 μ V ⁶ 0.029 % + 200 μ V ⁶ 0.077 % + 200 μ V ⁶ 0.29 % + 960 μ V ⁶ 0.96 % + 960 μ V ⁶	
(10 to 100) V	(20 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz	0.02 % + 3.9 mV ⁶ 0.02 % + 2.0 mV ⁶ 0.02 % + 2.0 mV ⁶ 0.034 % + 2.0 mV ⁶ 0.12 % + 2.0 mV ⁶	
(100 to 700) V	50 Hz to 1 kHz	0.039 % + 20 mV ⁶	
Linearity at 10V _p			
(-40 to 0) dB	20 Hz to 100 kHz (100 to 300) kHz 300 kHz to 1 MHz	0.005 dB 0.037 dB 0.12 dB	Multimeter
	100 kHz to 15 MHz	0.068 dB	Attenuator

Parameter/Equipment	Frequency	CMC ^{2,5} (±)	Comments
AC Current – Generate (2.2 to 22) mA (22 to 220) mA (0.22 to 1) A	50 Hz to 1 kHz 50 Hz to 1 kHz 50 Hz to 1 kHz	0.015 % + 0.36 µA 0.015 % + 3.6 µA 0.066 % + 36 µA	Calibrator
AC Current – Measure (1 to 10) mA (10 to 100) mA (0.1 to 1) A	(50 to 100) Hz 100 Hz to 1 kHz (50 to 100) Hz 100 Hz to 1 kHz (50 to 100) Hz 100 Hz to 1 kHz	0.07 % + 2.4 µA 0.035 % + 2.4 µA 0.07 % + 24 µA 0.035 % + 24 µA 0.077 % + 200 µA ⁶ 0.096 % + 200 µA ⁶	Multimeter
AC Phase – Measure (-180 to +180) deg.	10 Hz to 100 kHz 100 kHz to 1 MHz (1 to 5) MHz	0.02 ° 0.04 ° 0.12 °	Digital oscilloscope
AC Distortion – Measure	400 Hz, 1 kHz	0.12 %	Measuring receiver

III. Electrical – RF/Microwave

Parameter/Frequency	Range	CMC ^{2, 4, 5} ($\pm$)	Comments
RF Power – Generate ³			
50 MHz	1 mW	0.53 %	Reference oscillator
100 kHz to 5 GHz	0 dBm	0.05 dB	Power meter and power sensor
(5 to 12) GHz	0 dBm	0.07 dB	
(12 to 14) GHz	0 dBm	0.07 dB	
(14 to 18) GHz	0 dBm	0.09 dB	
(18 to 25) GHz	0 dBm	0.12 dB	
(25 to 40) GHz	0 dBm	0.15 dB	
100 kHz to 10 MHz	(-10 to 0) dBm (-100 to -10) dBm	0.06 dB 0.08 dB	Power meter and power sensor, measuring receiver, signal analyzer
(0.01 to 3) GHz	(0 to 20) dBm (-10 to 0) dBm (-100 to -10) dBm	0.09 dB 0.05 dB 0.08 dB	
(3 to 5) GHz	(0 to 20) dBm (-10 to 0) dBm (-100 to -10) dBm	0.12 dB 0.05 dB 0.07 dB	
(5 to 12) GHz	(0 to 20) dBm (-10 to 0) dBm (-100 to -10) dBm	0.16 dB 0.07 dB 0.08 dB	
(12 to 18) GHz	(0 to 20) dBm (-10 to 0) dBm (-100 to -10) dBm	0.16 dB 0.09 dB 0.11 dB	

Parameter/Frequency	Range	CMC ^{2, 4, 5} (±)	Comments
RF Power – Measuring Equipment ³			
Power Sensor Correction Factor			
100 kHz	0 dBm	1.6 %	Power meter and power sensor
300 kHz	0 dBm	1.1 %	
500 kHz	0 dBm	0.99 %	
1 MHz	0 dBm	0.98 %	
3 MHz	0 dBm	0.98 %	
5 MHz	0 dBm	0.98 %	
10 MHz	0 dBm	0.72 %	
30 MHz	0 dBm	0.72 %	
50 MHz	0 dBm	0.72 %	
100 MHz	0 dBm	0.72 %	
300 MHz	0 dBm	0.72 %	
500 MHz	0 dBm	0.71 %	
1 GHz	0 dBm	0.72 %	
1.5 GHz	0 dBm	1.2 %	
2 GHz	0 dBm	0.72 %	
2.5 GHz	0 dBm	1.3 %	
3 GHz	0 dBm	0.72 %	
4 GHz	0 dBm	0.72 %	
4.2 GHz	0 dBm	1.4 %	
5 GHz	0 dBm	0.73 %	
6 GHz	0 dBm	0.73 %	
7 GHz	0 dBm	0.86 %	
8 GHz	0 dBm	0.91 %	
9 GHz	0 dBm	0.99 %	
10 GHz	0 dBm	1.1 %	
11 GHz	0 dBm	1.1 %	
12 GHz	0 dBm	1.1 %	
12.4 GHz	0 dBm	1.3 %	
13 GHz	0 dBm	0.95 %	
14 GHz	0 dBm	1.2 %	
15 GHz	0 dBm	1.2 %	
16 GHz	0 dBm	1.2 %	
17 GHz	0 dBm	1.3 %	
18 GHz	0 dBm	1.3 %	
19 GHz	0 dBm	1.3 %	
20 GHz	0 dBm	1.5 %	
21 GHz	0 dBm	1.6 %	
22 GHz	0 dBm	1.7 %	
23 GHz	0 dBm	1.7 %	
24 GHz	0 dBm	1.9 %	
25 GHz	0 dBm	2.1 %	

Parameter/Frequency	Range	CMC ^{2,4,5} ($\pm$)	Comments
RF Power Measuring Equipment ³ (cont) –			
Power Sensor Correction Factor			
26 GHz	0 dBm	3.1 %	Power meter and power sensor
26.5 GHz	0 dBm	3.3 %	
27 GHz	0 dBm	3.5 %	
28 GHz	0 dBm	3.3 %	
29 GHz	0 dBm	3.0 %	
30 GHz	0 dBm	2.9 %	
31 GHz	0 dBm	2.9 %	
32 GHz	0 dBm	2.9 %	
33 GHz	0 dBm	2.8 %	
34 GHz	0 dBm	2.7 %	
35 GHz	0 dBm	2.6 %	
36 GHz	0 dBm	2.5 %	
37 GHz	0 dBm	2.6 %	
38 GHz	0 dBm	2.7 %	
39 GHz	0 dBm	2.8 %	
40 GHz	0 dBm	2.8 %	
(0.1 to 1) MHz	0 dBm	1.6 %	
1 MHz to 2 GHz	0 dBm	1.3 %	
(2 to 6) GHz	0 dBm	1.3 %	
(6 to 8) GHz	0 dBm	1.6 %	
(8 to 12) GHz	0 dBm	1.7 %	
(12 to 18) GHz	0 dBm	2.2 %	
Power Meter Linearity			
10 MHz to 6 GHz	(0 to 10) dBm	3.9 %	Power meter and power sensor, step attenuator
(6 to 12) GHz	(0 to 10) dBm	6.3 %	
(12 to 18) GHz	(0 to 10) dBm	7.0 %	
10 MHz to 6 GHz	(-50 to 0) dBm	4.6 %	
(6 to 12) GHz	(-50 to 0) dBm	6.6 %	
(12 to 18) GHz	(-50 to 0) dBm	7.0 %	

Parameter/Frequency	Range	CMC ^{2, 4, 5} ($\pm$)	Comments
Frequency Modulation – Measure (0.25 to 10) MHz Rate: 20 Hz to 10 kHz (10 to 1300) MHz Rate: 20 Hz to 200 kHz (10 to 1300) MHz Rate: 50 Hz to 100 kHz	Deviation: (0.3 to 4) kHz Deviation: (4 to 40) kHz Deviation: (0.3 to 4) kHz Deviation: (4 to 40) kHz Deviation: (40 to 400) kHz Deviation: (0.3 to 4) kHz Deviation: (4 to 40) kHz Deviation: (40 to 400) kHz	2.5 % + 1 Hz 2.5 % + 10 Hz 5.9 % + 1 Hz 5.9 % + 10 Hz 5.9 % + 100 Hz 1.3 % + 1 Hz 1.3 % + 10 Hz 1.3 % + 100 Hz	Measuring receiver
Frequency Modulation – Measuring Equipment (0.25 to 10) MHz Rate: 20 Hz to 10 kHz (10 to 1300) MHz Rate: 20 Hz to 200 kHz (10 to 1300) MHz Rate: 50 Hz to 100 kHz	Deviation: (0.3 to 4) kHz Deviation: (4 to 40) kHz Deviation: (0.3 to 4) kHz Deviation: (4 to 40) kHz Deviation: (40 to 400) kHz Deviation: (0.3 to 4) kHz Deviation: (4 to 40) kHz Deviation: (40 to 400) kHz	2.4 % + 1 Hz 2.4 % + 10 Hz 5.9 % + 1 Hz 5.9 % + 10 Hz 5.9 % + 100 Hz 1.3 % + 1 Hz 1.3 % + 10 Hz 1.3 % + 100 Hz	Measuring receiver
Attenuation – Measure 10 MHz to 18 GHz	(0 to 90) dB (90 to 100) dB	0.06 dB 0.08 dB	Measuring receiver
Attenuation - Measuring Equipment 10 MHz to 12 GHz (12 to 18) GHz	(0 to 100) dB (0 to 100) dB	0.19 dB 0.32 dB	Step attenuator

Parameter/Equipment	Range	CMC ^{2, 4, 5} (±)	Comments
Amplitude Modulation – Measure			
(0.15 to 10) MHz Rate: 20 Hz to 10 kHz	Depth: (5 to 40) % Depth: (40 to 99) %	3.6 % + 0.01 % of full scale 3.6 % + 0.1 % of full scale	Measuring receiver
(0.15 to 10) MHz Rate: 50 Hz to 10 kHz	Depth: (5 to 40) % Depth: (40 to 99) %	2.4 % + 0.01 % of full scale 2.4 % + 0.1 % of full scale	
(10 to 1300) MHz Rate: 20 Hz to 100 kHz	Depth: (5 to 40) % Depth: (40 to 99) %	3.6 % + 0.01 % of full scale 3.6 % + 0.1 % of full scale	
(10 to 1300) MHz Rate: 50 Hz to 50 kHz	Depth: (5 to 40) % Depth: (40 to 99) %	1.3 % + 0.01 % of full scale 1.3 % + 0.1 % of full scale	
Amplitude Modulation – Measuring Equipment			
(0.15 to 10) MHz Rate: 20 Hz to 10 kHz	Depth: (5 to 40) % Depth: (40 to 99) %	3.6 % + 0.01 % of full scale 3.6 % + 0.1 % of full scale	Measuring receiver
(0.15 to 10) MHz Rate: 50 Hz to 10 kHz	Depth: (5 to 40) % Depth: (40 to 99) %	2.4 % + 0.01 % of full scale 2.4 % + 0.1 % of full scale	
(10 to 1300) MHz Rate: 20 Hz to 100 kHz	Depth: (5 to 40) % Depth: (40 to 99) %	3.6 % + 0.01 % of full scale 3.6 % + 0.1 % of full scale	
(10 to 1300) MHz Rate: 50 Hz to 50 kHz	Depth: (5 to 40) % Depth: (40 to 99) %	1.3 % + 0.01 % of full scale 1.3 % + 0.1 % of full scale	

Parameter/Equipment	Range	CMC ^{2, 4, 5} ($\pm$)	Comments
Spectrum Analyzer (cont) ⁸ –			
Frequency Response, Reference Level Accuracy, Absolute Amplitude Accuracy	21 GHz 22 GHz 23 GHz 24 GHz 25 GHz 26 GHz 26.5 GHz 27 GHz 28 GHz 29 GHz 30 GHz 31 GHz 32 GHz 33 GHz 34 GHz 35 GHz 36 GHz 37 GHz 38 GHz 39 GHz 40 GHz (0.1 to 1) MHz 1 MHz to 2 GHz (2 to 6) GHz (6 to 8) GHz (8 to 12) GHz (12 to 18) GHz	0.15 dB 0.14 dB 0.15 dB 0.16 dB 0.17 dB 0.21 dB 0.21 dB 0.21 dB 0.21 dB 0.21 dB 0.22 dB 0.24 dB 0.22 dB 0.2 dB 0.19 dB 0.18 dB 0.18 dB 0.19 dB 0.18 dB 0.19 dB 0.2 dB 0.11 dB 0.1 dB 0.12 dB 0.13 dB 0.14 dB 0.17 dB	Power meter and power sensor
Reference Level Accuracy, Absolute Amplitude Accuracy			
(0.01 to 2) GHz (2 to 10) GHz (10 to 12) GHz (12 to 18) GHz	(-100 to -10) dBm (-100 to -10) dBm (-100 to -10) dBm (-100 to -10) dBm	0.1 dB 0.16 dB 0.16 dB 0.28 dB	Power meter and power sensor, step attenuator
Attenuator Switching Accuracy	10 kHz to 40 GHz	0.09 dB	Signal generator
Bandwidth Switching Accuracy	10 kHz to 40 GHz	0.09 dB	
Bandwidth Accuracy	10 Hz to 50 MHz	3.5 %	
Scale Fidelity – (0.01 to 18) GHz	(10 to 90) dB (90 to 100) dB	0.07 dB 0.09 dB	Step Attenuator

Parameter/Equipment	Range	CMC ^{2,4} (±)	Comments
Signal Generator –			
Standard Frequency Accuracy	10 MHz	3.5 µHz	Cesium standard and precision phase comparator
Frequency Accuracy	10 Hz to 1 MHz (1 to 10) MHz 10 MHz 10 MHz to 1 GHz (1 to 10) GHz (10 to 20) GHz (20 to 30) GHz (30 to 40) GHz	0.012 Hz 0.85 mHz 0.85 mHz 0.075 Hz 0.14 Hz 0.26 Hz 0.37 Hz 0.49 Hz	Cesium standard and frequency counter
Linearity			
(0.01 to 18) GHz	(-50 to 20) dBm (-100 to -50) dBm	0.28 dB 0.37 dB	Spectrum analyzer and step attenuator
Absolute Level Accuracy			
100 kHz to 5 GHz	0 dBm	0.05 dB	Power meter and power sensor
(5 to 12) GHz	0 dBm	0.07 dB	
(12 to 14) GHz	0 dBm	0.07 dB	
(14 to 18) GHz	0 dBm	0.09 dB	
(18 to 25) GHz	0 dBm	0.12 dB	
(25 to 40) GHz	0 dBm	0.15 dB	
100 kHz to 10 MHz	(-10 to 0) dBm (-100 to -10) dBm	0.06 dB 0.08 dB	Power meter and power sensor, measuring receiver
(0.01 to 3) GHz	(0 to 20) dBm (-10 to 0) dBm (-100 to -10) dBm	0.09 dB 0.05 dB 0.08 dB	
(3 to 5) GHz	(0 to 20) dBm (-10 to 0) dBm (-100 to -10) dBm	0.12 dB 0.05 dB 0.07 dB	

Parameter/Equipment	Range	CMC ^{2, 4, 5} ($\pm$)	Comments
Signal Generator – (cont)			
Absolute Level Accuracy			
(5 to 12) GHz	(0 to 20) dBm (-10 to 0) dBm (-100 to -10) dBm	0.16 dB 0.07 dB 0.08 dB	Power meter and power sensor, measuring receiver
(12 to 18) GHz	(0 to 20) dBm (-10 to 0) dBm (-100 to -10) dBm	0.16 dB 0.09 dB 0.11 dB	
2nd and 3rd Harmonic			
(0.02 to 3) GHz	(0 to -70) dBc	0.9 dB	
(3 to 13.5) GHz	(0 to -90) dBc	1.7 dB	Signal Analyzer
(13.5 to 18) GHz	(0 to -90) dBc	2.9 dB	
FM Modulation Accuracy			
(0.25 to 10) MHz Mod. Frequency: 20 Hz to 10 kHz	Deviation: (0.3 to 4) kHz Deviation: (4 to 40) kHz	2.5 % + 1 Hz 2.5 % + 10 Hz	Measuring receiver
(10 to 1300) MHz Mod. Frequency: 20 Hz to 200 kHz	Deviation: (0.3 to 4) kHz Deviation: (4 to 40) kHz Deviation: (40 to 400) kHz	5.9 % + 1 Hz 5.9 % + 10 Hz 5.9 % + 100 Hz	
(10 to 1300) MHz Mod. Frequency: 50 Hz to 100 kHz	Deviation: (0.3 to 4) kHz Deviation: (4 to 40) kHz Deviation: (40 to 400) kHz	1.3 % + 1 Hz 1.3 % + 10 Hz 1.3 % + 100 Hz	

Parameter/Equipment	Range	CMC ^{2, 4, 5} (±)	Comments
Signal Generator – (cont)			
AM Modulation Accuracy			
(0.15 to 10) MHz Rate: 20 Hz to 10 kHz	Depth: (5 to 40) % Depth: (40 to 99) %	3.6 % + 0.01 % of full scale 3.6 % + 0.1 % of full scale	Measuring receiver
(0.15 to 10) MHz Rate: 50 Hz to 10 kHz	Depth: (5 to 40) % Depth: (40 to 99) %	2.4 % + 0.01 % of full scale 2.4 % + 0.1 % of full scale	
(10 to 1300) MHz Rate: 20 Hz to 100 kHz	Depth: (5 to 40) % Depth: (40 to 99) %	3.6 % + 0.01 % of full scale 3.6 % + 0.1 % of full scale	
(10 to 1300) MHz Rate: 50 Hz to 50 kHz	Depth: (5 to 40) % Depth: (40 to 99) %	1.3 % + 0.01 % of full scale 1.3 % + 0.1 % of full scale	
Attenuator – Attenuation Accuracy			
10 MHz to 12 GHz	(0 to 90) dB (90 to 100) dB	0.06 dB 0.08 dB	Measuring receiver
(12 to 18) GHz	(0 to 90) dB (90 to 100) dB	0.06 dB 0.08 dB	

Parameter/Frequency	Range	CMC ² ($\pm$)	Comments/ Procedure
S Parameters – S_{11} , S_{12} , S_{21} , S_{22} Reflection S_{11}/S_{22} Magnitude and Phase – Measure ⁹			
(70 to 300) kHz	(0.0001 to 0.03) lin	(0.005 to 0.011) lin (90 to 13) deg	VNA: MS4644A calibration kit : 3652A, 3652A-1, 3653, 3653A Verification kit: 3668-1
	(0.03 to 0.1) lin	(0.005 to 0.011) lin (13 to 4.2) deg	
	(0.1 to 0.2) lin	(0.005 to 0.011) lin (4.2 to 2.3) deg	
	(0.2 to 0.5) lin	(0.005 to 0.011) lin (2.3 to 0.82) deg	
	(0.5 to 1) lin	(0.008 to 0.021) lin (2.3 to 0.82) deg	
300 kHz to 18 GHz	(0.0001 to 0.03) lin	(0.005 to 0.02) lin (90 to 21) deg	
	(0.03 to 0.1) lin	(0.005 to 0.02) lin (21 to 7.4) deg	
	(0.1 to 0.2) lin	(0.005 to 0.02) lin (7.4 to 4.5) deg	
	(0.2 to 0.5) lin	(0.005 to 0.02) lin (4.5 to 0.80) deg	
	(0.5 to 1) lin	(0.008 to 0.041) lin (4.5 to 0.80) deg	
(18 to 40) GHz	(0.0001 to 0.03) lin	(0.011 to 0.033) lin (90 to 21) deg	
	(0.03 to 0.1) lin	(0.011 to 0.033) lin (90 to 21) deg	
	(0.1 to 0.2) lin	(0.011 to 0.033) lin (21 to 7.4) deg	

Parameter/Frequency	Range	CMC ² ($\pm$)	Comments/ Procedure
S Parameters – S_{11} , S_{12} , S_{21} , S_{22} (cont)			
Reflection S_{11}/S_{22} Magnitude and Phase – Measure ⁹			
(18 to 40) GHz	(0.2 to 0.5) lin	(0.011 to 0.033) lin (4.5 to 1.2) deg	VNA: MS4644A calibration kit : 3652A, 3652A-1, 3653, 3653A Verification kit: 3668-1
	(0.5 to 1) lin	(0.020 to 0.059) lin (4.5 to 1.2) deg	
Transmission S_{12}/S_{21} , Magnitude and Phase – Measure ⁹			
(70 to 300) kHz	(0 to -5) dB	(0.062 to 0.019) dB (0.41 to 0.13) deg	
	(-5 to -30) dB	(0.019 to 0.039) dB (0.13 to 0.26) deg	
	(-30 to -35) dB	(0.019 to 0.044) dB (0.13 to 0.30) deg	
	(-35 to -60) dB	(0.024 to 0.045) dB (0.16 to 0.30) deg	
	(-60 to -70) dB	(0.025 to 0.048) dB (0.17 to 0.32) deg	
	(-70 to -80) dB	(0.034 to 0.11) dB (0.23 to 0.67) deg	
300 kHz to 18 GHz	(0 to -5) dB	(0.19 to 0.017) dB (1.8 to 0.11) deg	
	(-5 to -30) dB	(0.017 to 0.20) dB (0.11 to 1.7) deg	
	(-30 to -35) dB	(0.017 to 0.20) dB (0.11 to 1.7) deg	

Parameter/Frequency	Range	CMC ² (±)	Comments/ Procedure
S Parameters – S ₁₁ , S ₁₂ , S ₂₁ , S ₂₂ (cont)			
Transmission S ₁₂ /S ₂₁ , Magnitude and Phase – Measure ⁹			
300 kHz to 18 GHz (18 to 40) GHz	(-35 to -60) dB	(0.022 to 0.20) dB (0.15 to 1.7) deg	VNA: MS4644A calibration kit : 3652A, 3652A-1, 3653, 3653A Verification kit: 3668-1
	(-60 to -70) dB	(0.023 to 0.20) dB (0.16 to 1.7) deg	
	(-70 to -80) dB	(0.033 to 0.20) dB (0.22 to 1.9) deg	
	(0 to -5) dB	(0.19 to 0.16) dB (1.8 to 1.7) deg	
	(-5 to -30) dB	(0.039 to 0.31) dB (0.26 to 3.3) deg	
	(-30 to -35) dB	(0.039 to 0.31) dB (0.26 to 3.4) deg	
	(-35 to -60) dB	(0.044 to 0.31) dB (0.30 to 3.4) deg	
	(-60 to -70) dB	(0.044 to 0.31) dB (0.30 to 3.4) deg	
	(-70 to -80) dB	(0.044 to 0.31) dB (0.30 to 3.4) deg	

IV. Time & Frequency

Parameter/Equipment	Frequency	CMC ^{2,4} ($\pm$)	Comments
Frequency – Measure			
High Stability Frequency Standard	10 MHz	3.5 μ Hz	Cesium standard and precision phase comparator
Frequency Generators	10 Hz to 1 MHz (1 to 10) MHz 10 MHz 10 MHz to 1 GHz (1 to 10) GHz (10 to 20) GHz (20 to 30) GHz (30 to 40) GHz	0.012 Hz 0.85 mHz 0.85 mHz 0.075 Hz 0.14 Hz 0.26 Hz 0.37 Hz 0.49 Hz	Cesium standard and frequency counter
Frequency ³ – Measure	10 Hz to 1 MHz (1 to 10) MHz (10 to 100) MHz 100 MHz to 1 GHz (1 to 10) GHz (10 to 20) GHz (20 to 30) GHz (30 to 40) GHz	0.012 Hz 0.02 Hz 0.2 Hz 2 Hz 20 Hz 41 Hz 61 Hz 80 Hz	Rubidium standard and frequency counter
Frequency – Measuring Equipment	10 Hz to 1 MHz (1 to 10) MHz 10 MHz to 1 GHz	0.058 Hz 0.058 Hz 0.17 Hz	Cesium standard and signal generator
Counters and Timers	(1 to 10) GHz (10 to 20) GHz (20 to 30) GHz (30 to 40) GHz 10 MHz	0.22 Hz 0.37 Hz 0.49 Hz 0.63 Hz 0.067 mHz	Cesium standard

¹ This laboratory offers commercial calibration service.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ Field calibration service is available for this calibration and this laboratory meets A2LAR104 – *General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the Calibration and Measurement Capabilities Uncertainty (CMC) found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

⁴ Uncertainty does not include mismatch error due to connections of the device to other devices in actual use. Mismatch uncertainties, due to the reflection coefficient of the device to be calibrated, are to be included in the overall measurement uncertainty. The approach of determining expanded uncertainties at approximately the 95% level of confidence, (using a coverage factor of $k = 2$) is to be applied for this calculation as well.

⁵ Unless otherwise defined, in the statement of CMC, the value of % is defined as the percentage of reading.

⁶ The coverage factor of $k = 1.65$ is used such that the level of confidence corresponds to approximately 95 %.

⁷ xx are suffix letters A to Z or number 0 to 9 for LTE RF conformance test system options.

⁸ Includes other frequency selective level measuring equipment like Signal Analyzers

⁹ CMC for intermediate values of measurand can be found by interpolation



Accredited Laboratory

A2LA has accredited

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for technical competence in the field of

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This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSLI Z540-1-1994 and any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (*refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009*).



Presented this 29th day of June 2015.

A handwritten signature in blue ink, appearing to read 'J. C. Bunt', written over a horizontal line.

Senior Director of Quality & Communications
For the Accreditation Council
Certificate Number 3128.01
Valid to May 31, 2017
Revised October 27, 2016

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.